QP 25901272

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFTQ201 - FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutesMaximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Part A

Short Answer type

Answer any 5 questions

Each question carries 2 marks

1. Where do colonies grow in pour plate?[U] / [CO1]

Answer: Colonies grow both on the surface and within the depth of the agar medium.

2. State one limitation of continuous culture systems.[K] / [CO3]

Answer: High risk of contamination during prolonged operation.

3. What toxin is produced by *Shigelladysenteriae*?[U] / [CO4]

Answer: Shiga toxin.

4. Example of differential media.[U] / [CO1]
Answer: MacConkey agar.

5. What is the effect of temperature on microbial growth?[K] / [CO3]

Answer: Temperature affects enzyme activity and membrane fluidity, influencing growth rate; extreme temperatures inhibit or kill microorganisms.

6. What is phenol used for?[E] / [CO1]

Answer: Phenol is used as a disinfectant and antiseptic.

7. Define compound microscope.[U] / [CO1]

Answer: A compound microscope is an optical instrument that uses two lenses (objective and eyepiece) to magnify small objects.

Part B

Short Essay Questions

Answer any 5 questions

Each question carries 4 marks

8. Outline the laboratory methods used for isolation and identification of *Listeria monocytogenes* from food and clinical samples.[K] / [CO4]
Answer:

- Enrichment in selective broth such as Listeria enrichment broth
- Cold enrichment technique
- Plating on selective media such as PALCAM or Oxford agar
- Identification by Gram staining (Gram positive rods)
- Biochemical tests such as catalase positive and motility at 25°C
- CAMP test showing enhanced hemolysis
- Molecular methods such as PCR for confirmation

9. Classify bacteria based on their shape and arrangement with examples.[U] / [CO3]
Answer:

- Cocci: spherical (Staphylococcus, Streptococcus)
- Bacilli: rod shaped (Escherichia coli)
- Spirilla: spiral shaped (Spirillum)
  Arrangements:
- Diplococci (Neisseria)
- Streptococci (chains)
- Staphylococci (clusters)

10. Working of fluorescent microscope.[U] / [CO1]
Answer:

- Uses UV or blue light to excite fluorochromes
- Fluorochromes emit visible light of longer wavelength
- Filters allow only emitted light to reach the eye
- Used to detect specific structures or microorganisms

11. Outline the pathogenesis of *Bacillus cereus* food intoxication.
[U] / [CO1]
Answer:

- Ingestion of contaminated food
- Production of toxins
    - Emetic toxin (heat stable) causes vomiting
    - Diarrheal toxin (heat labile) affects intestine
- Toxins disrupt epithelial cells leading to symptoms

12. Explain the significance of bacterial arrangements in identifying bacterial species.
[U] / [CO3]
Answer:

- Arrangement helps in preliminary identification
- Reflects mode of cell division
- Distinguishes genera such as Streptococcus and Staphylococcus
- Useful in microscopy based diagnosis

13. Explain the pathogenesis of enterohemorrhagic*E. coli* (EHEC) infection.[K] /[CO4]
Answer:

- Ingestion of contaminated food
- Colonization of intestinal epithelium
- Production of Shiga like toxin
- Inhibition of protein synthesis
- Causes hemorrhagic colitis and hemolytic uremic syndrome

14. Explain Koch's postulates with applications.[U] / [CO1]
Answer:

- Microorganism must be present in diseased host
- Must be isolated and grown in pure culture
- Should reproduce disease in healthy host
- Must be re-isolated from infected host
  Applications: Establish causative agents of infectious diseases

Part C

Long Essay Questions

Answer any 2 questions

Each question carries 10 marks

15. Compare and contrast staphylococcal food intoxication with food infections caused by *Salmonella* and *Clostridium perfringens*.[K] / [CO1]

Answer:

- Staphylococcus aureus
    - Preformed toxin
    - Short incubation period
    - Vomiting predominant
- Salmonella
    - Infection after ingestion
    - Invades intestinal mucosa
    - Causes diarrhea and fever
- Clostridium perfringens
    - Spore forming
    - Toxin produced in intestine
    - Causes abdominal cramps and diarrhea
- Key differences include toxin type, incubation time, and mechanism

16. Detailed comparison of TEM and SEM.[U] / [CO1, CO2]
Answer:

- TEM
    - Uses transmitted electrons
    - High resolution
    - Internal structure observed
- SEM
    - Uses reflected electrons
    - 3D surface image
    - Lower resolution than TEM
- TEM requires thin sections, SEM requires surface coating
- Applications differ in ultrastructure vs surface morphology

17. Explain the prevention and control measures of *Clostridium perfringens* food infection in food handling and storage.[An] / [CO4]
Answer:

- Proper cooking of food
- Rapid cooling after cooking
- Avoid prolonged storage at room temperature
- Reheating food adequately
- Maintain hygiene during food handling
- Prevent cross contamination

18. Explain the prevention and control measures of salmonellosis at household and community levels. [U] / [CO4]
Answer:

- Proper cooking of poultry and eggs
- Avoid raw or undercooked food
- Maintain kitchen hygiene
- Hand washing practices
- Safe water supply
- Public health surveillance
- Food safety education